# Notes: Welch & Goyal (RFS, 2008)

## A Comprehensive Look at the Empirical Performance of Equity Premium Prediction

## Contents


**1 Big picture** **2**

**2 Data and measurement** **2**

2.1 Dependent variable: the equity premium . . . 2

2.2 Predictor variables . . . 3

2.3 In-sample versus out-of-sample forecasts . . . 3

2.4 Forecast-error statistics . . . 3

2.5 Bootstrap inference . . . 4

2.6 Campbell–Thompson restrictions . . . 4

**3 Empirical design and credibility** **4**

3.1 What the paper is testing . . . 4

3.2 Why out-of-sample tests are central . . . 4

3.3 What makes the exercise comprehensive . . . 5

3.4 Limits of the exercise . . . 5

**4 Tables and figures** **5**

4.1 Figures 1–2 and Table 1: annual forecasts . . . 5

4.2 Table 2: five-year forecasts . . . 10

4.3 Table 3 and Figure 3: monthly forecasts with restrictions . . . 10

4.4 Tables 4–5: smoothed ratios and persistence corrections . . . 12

4.5 Table 6: encompassing tests . . . 13

**5 Short summary** **13**

**6 Conclusion** **14**

6.1 Core conclusion . . . 14

6.2 Economic intuition . . . 14

6.3 Takeaway . . . 14

# 1 Big picture

- **Question.** Do the predictive variables proposed in the return-predictability literature reliably forecast the aggregate equity premium?
- **Core answer.** Mostly no. Many predictors look impressive in earlier in-sample tests, but their performance is unstable and generally poor out of sample.
- **Main empirical setting.** The paper compares a large set of predictor variables using the same data construction, forecast windows, test statistics, bootstrap critical values, and investment-oriented diagnostics.
- **Benchmark.** The benchmark forecast is the historical average equity premium. A useful predictive model should beat this simple unconditional forecast out of sample.
- **Key annual result.** At annual frequency, most models have negative out-of-sample $R^2$ values. Even variables with positive in-sample $R^2$ often fail when forecasts are made recursively with only past data.
- **Instability result.** Much of the historical success of valuation ratios and other predictors comes from specific episodes, especially the 1973–1975 Oil Shock and earlier subsamples.
- **Long-horizon result.** Five-year overlapping return forecasts do not rescue the literature. Some long-horizon predictors appear significant in sample, but out-of-sample performance remains fragile and often negative.
- **Monthly result.** Some Campbell–Thompson-style restrictions improve monthly forecasts, but the improvements are small, model-dependent, and not enough to overturn the broader conclusion.
- **Market timing result.** The models would not have reliably helped an investor with access only to information available at the forecast date to time the market profitably.
- **Economic takeaway.** Equity premium predictability is much weaker and less stable than the earlier literature suggested. The correct default position is skepticism toward aggregate return-forecasting variables unless they pass strict out-of-sample tests.

# 2 Data and measurement

## 2.1 Dependent variable: the equity premium

The dependent variable is the aggregate equity premium:

$$r_t^e = r_t^m - r_t^f, \tag{1}$$

where $r_t^m$ is the stock market return and $r_t^f$ is the short-term risk-free rate.

- Monthly stock returns are S&P 500 returns from CRSP for 1926–2005.
- Annual and longer-horizon stock market data can begin in 1871 using Robert Shiller's data.
- From 1920 onward, the risk-free rate is the Treasury-bill rate.
- Before 1920, the risk-free rate is imputed from commercial paper rates using

$$\text{Treasury bill rate} = -0.004 + 0.886 \times \text{commercial paper rate}.$$

- The paper studies annual, five-year overlapping, and monthly equity premia.

**Interpretation:** The paper is not asking whether a particular return series can be fit ex post. It asks whether investors could have forecasted the equity premium using variables known at the forecast date.

## 2.2 Predictor variables

The paper evaluates the main variables proposed in the return-predictability literature:

- **Valuation ratios:** dividend-price ratio $d/p$, dividend yield $d/y$, earnings-price ratio $e/p$, dividend-payout ratio $d/e$, and smoothed earnings or dividend ratios such as $e^{10}/p$.
- **Risk and volatility:** stock variance $svar$ and cross-sectional premium $csp$.
- **Corporate finance variables:** book-to-market $b/m$, net equity expansion $ntis$, percent equity issuing $eqis$, and investment-capital ratio $i/k$.
- **Interest rates and spreads:** Treasury-bill rate $tbl$, long-term yield $lty$, long-term return $ltr$, term spread $tms$, default yield spread $dfy$, and default return spread $dfr$.
- **Macro variables:** inflation $infl$ and consumption-wealth-income variables $cayp$ and $caya$.
- **Combination models:** kitchen-sink model $all$ and model-selection method $ms$.

**Interpretation:** The comparison is deliberately broad. It covers standard valuation ratios, interest-rate variables, corporate issuance measures, macro predictors, and combined models.

## 2.3 In-sample versus out-of-sample forecasts

The basic predictive regression is

$$r_t^e = \gamma_0 + \gamma_1 x_{t-1} + \varepsilon_t. \tag{1}$$

In-sample tests estimate the regression over a full sample and ask whether $\gamma_1$ is statistically significant. Out-of-sample tests estimate the model recursively and forecast using only information known before the return is realized.

**Interpretation:** The paper treats out-of-sample evidence as both a model diagnostic and as the economically relevant experiment for a real investor.

## 2.4 Forecast-error statistics

Let $e_N$ be the forecast error from the historical mean benchmark and $e_A$ be the error from the predictive model. Let $MSE_N$ and $MSE_A$ be their mean squared errors.

The out-of-sample $R^2$ is

$$R^2_{OOS} = 1 - \frac{MSE_A}{MSE_N}. \tag{2}$$

The adjusted version subtracts a degrees-of-freedom penalty:

$$\bar{R}^2_{OOS} = R^2_{OOS} - (1 - R^2_{OOS})\frac{T-k}{T-1}. \tag{3}$$

The root-MSE difference is

$$\Delta RMSE = \sqrt{MSE_N} - \sqrt{MSE_A}. \tag{4}$$

Positive $\Delta RMSE$ means the predictive model has lower forecast error than the historical average.

The McCracken-style statistic is

$$MSE\text{-}F = (T-h+1)\frac{MSE_N - MSE_A}{MSE_A}, \tag{5}$$

where $h$ is the horizon overlap.

**Interpretation:** A positive $R^2_{OOS}$ or $\Delta RMSE$ is economically intuitive: the conditional model beats the historical mean. A negative value means the predictor makes forecasts worse.

### 2.5 Bootstrap inference

- In-sample significance uses bootstrapped critical values for $F$-statistics.
- Out-of-sample significance uses one-sided tests because the null is that the conditional forecast is not better than the unconditional benchmark.
- Bootstrap procedures impose the null of no predictability and preserve properties such as serial correlation in predictors.
- Five-year forecasts are overlapping, so bootstrap inference is especially important.

**Interpretation:** The paper's conservative inference matters because many predictors are persistent and because standard asymptotic tests can overstate predictive significance.

### 2.6 Campbell–Thompson restrictions

For monthly data, the paper studies Campbell and Thompson style restrictions:

- coefficient sign restrictions based on theory;
- nonnegative equity-premium forecasts;
- restricted trading rules that prevent extreme negative or leveraged market positions.

**Interpretation:** These restrictions can improve noisy forecasts, but they also impose economic structure. The paper asks whether these restrictions rescue predictability in practice.

## 3 Empirical design and credibility

### 3.1 What the paper is testing

- The paper tests whether proposed predictors beat the historical average equity-premium forecast.
- The main experiment is recursive: estimate with data available up to time $t-1$, predict $t$, then expand the estimation window.
- Tests are run at annual, five-year, and monthly frequencies.
- Results are compared across the full available sample and important subsamples, especially forecasts beginning after 1965.

**Interpretation:** The design emphasizes the forecast problem faced by a real-time investor rather than the ex-post fit of a researcher.

### 3.2 Why out-of-sample tests are central

- In-sample significance can arise from data mining, persistence, or one unusual episode.
- Out-of-sample forecasting forces the model to keep working after its historical pattern is discovered.
- A model that only works early in the sample but fails later is not useful for investors and may be unstable or spurious.

**Interpretation:** The historical mean is hard to beat because the equity premium is noisy and expected returns are difficult to estimate from limited data.

### 3.3 What makes the exercise comprehensive

- The paper uses a large common dataset of predictors and returns.
- It evaluates variables under the same procedures rather than cherry-picking periods or methods.
- It includes valuation ratios, issuance variables, macro variables, interest-rate variables, and model combinations.
- It examines both statistical forecast performance and utility-style market timing performance.

**Interpretation:** The contribution is not a new predictor. The contribution is a disciplined horse race among many existing predictors.

### 3.4 Limits of the exercise

- The paper does not prove that expected returns are constant.
- A variable could have weak predictive power that is too small to detect reliably in finite samples.
- Forecast performance may depend on model form, data revisions, or economic restrictions not considered here.
- Later data after 2005 could change some conclusions, but the paper's main point is about robustness and real-time performance through the sample available to the authors.

**Interpretation:** The paper's conclusion is skeptical rather than absolutist: predictability may exist, but the published variables do not provide robust practical forecasts by 2005.

## 4 Tables and figures

The original paper's visuals are large multi-panel figures and wide tables. They are grouped below to save space while preserving the original content.

### 4.1 Figures 1–2 and Table 1: annual forecasts

**Read: Figure 1:**

- Annual predictors that are not in-sample significant generally show unstable cumulative forecast performance.
- Dividend-price and dividend-yield ratios perform well in some historical intervals but poorly after the Oil Shock and in the late sample.
- The red Oil Shock line highlights how much of the apparent predictability comes from one unusual episode.
- Negative or declining OOS cumulative SSE differences mean the historical mean forecast is doing better.

**Read: Figure 2:**

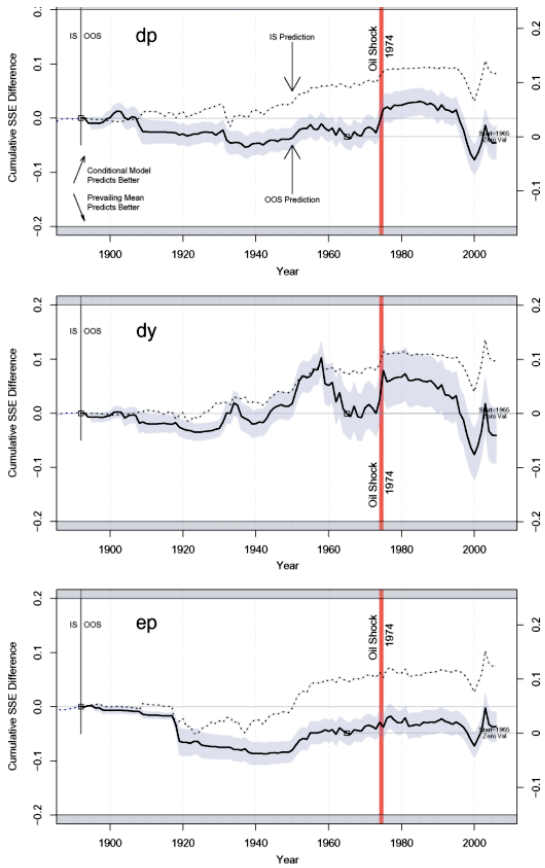
- Even predictors that are in-sample significant do not generally deliver robust annual OOS gains.
- Book-to-market performs well in an early subsample but loses performance after the Oil Shock and late in the sample.

- Net equity expansion and model selection perform poorly out of sample.
- The ex-post *cayp* variable looks strong, but its construction uses information not available in real time; the real-time *caya* version performs worse.

**Read: Table 1:**

- Many full-sample annual predictors are not significant in sample; most have negative OOS $R^2$.
- Among full-sample significant predictors, book-to-market, investment-capital, net equity expansion, percent equity issuing, and the kitchen-sink model are significant in sample.
- Significance in sample does not imply OOS success: for example, $b/m$ has negative OOS $R^2$ in the main annual OOS windows.
- Percent equity issuing is one of the few annual variables with positive full-sample OOS performance, but the result is not robust enough to overturn the broader pattern.
- The kitchen-sink model performs extremely badly OOS, illustrating the danger of overfitting.

**Economic message:** Annual return predictability largely fails the real-time forecast test. The problem is not only low statistical power; many conditional forecasts are worse than the historical mean.




ure 1
nual performance of IS insignificant predictors.

Figure 1, panels $d/p$, $d/y$, $e/p$

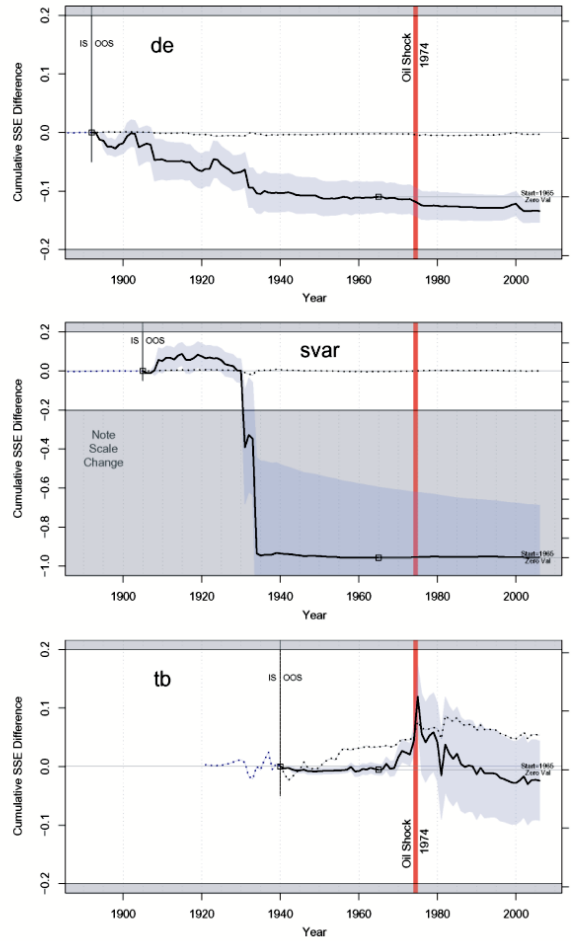



Figure 1, panels $d/e$, $svar$, $tbl$

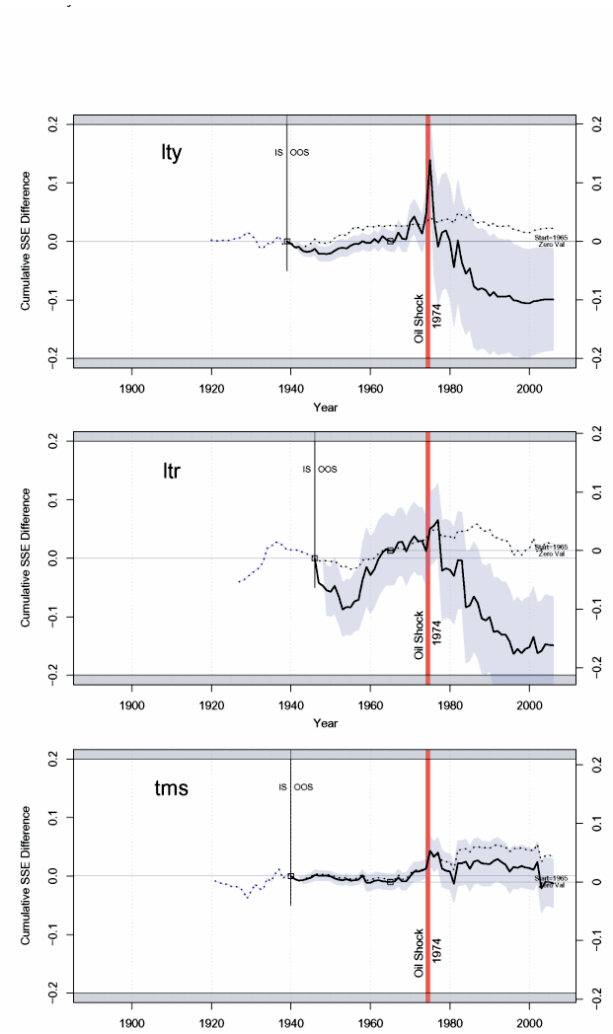



Figure 1, panels $lty$, $ltr$, $tms$

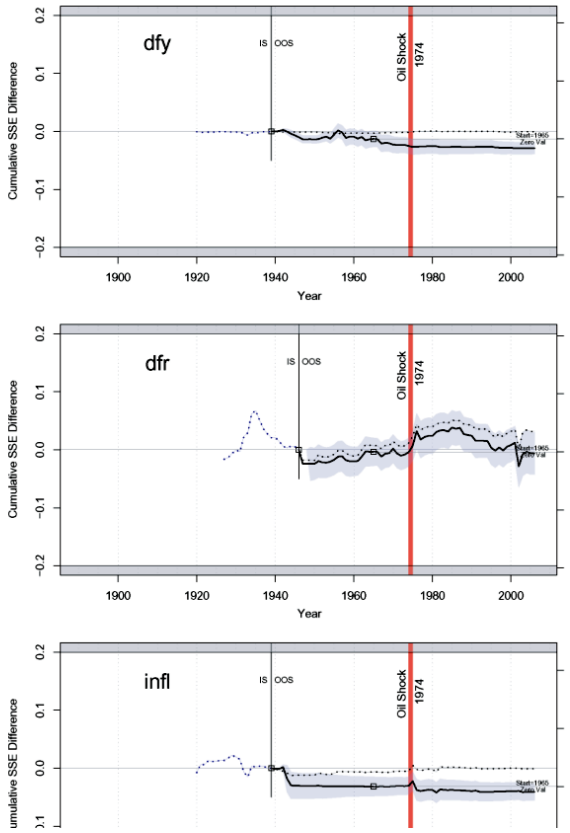
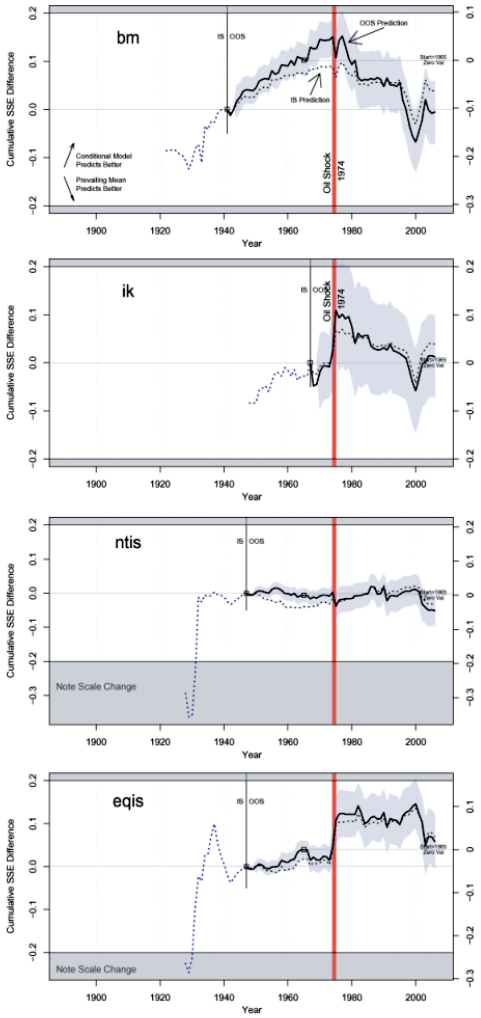



Figure 1, panels $dfy$, $dfr$, $infl$




ıre 2
**ual performance of predictors that are not in-sample significant**
lanation: See Figure 1.

Figure 2, panels $b/m$, $i/k$, $ntis$, $eqis$

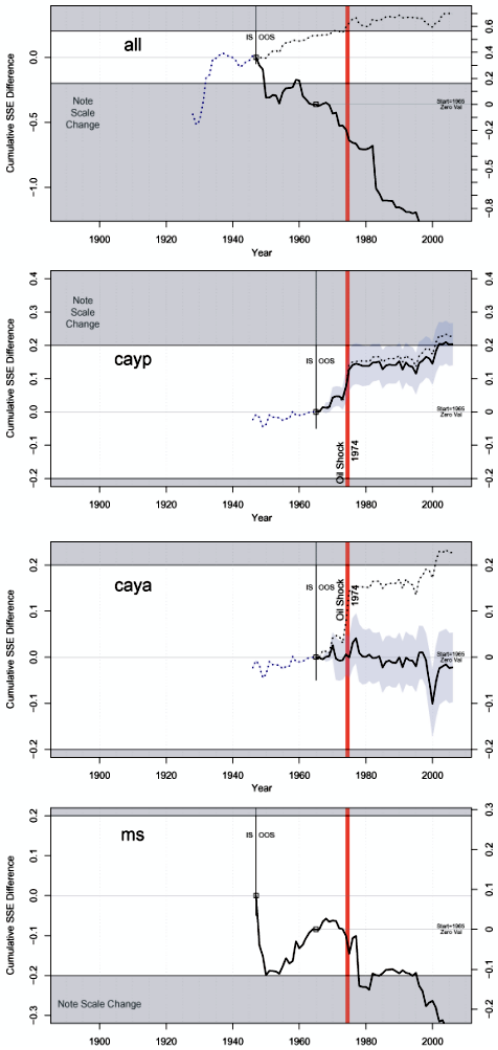



Figure 2, panels $all$, $cayp$, $caya$, $ms$

**Table 1**
**Forecasts at annual frequency**
**This table presents statistics on forecast errors in-sample (IS) and out-of-sample (OOS) for log equity premium forecasts at annual frequen equation and forecast). Variables are explained in Section 2. Stock returns are price changes, including dividends, of the S&P500. All numbe except except $\overline{R}^2$ and power which are simple percentages. A star next to IS-$\overline{R}^2$ denotes significance of the in-sample regression as measur values of which are obtained empirically from bootstrapped distributions). The column 'IS for OOS' gives the IS-$\overline{R}^2$ for the OOS perio (root mean square error) difference between the unconditional forecast and the conditional forecast for the same sample/forecast period. superior out-of-sample conditional forecast. The OOS-$\overline{R}^2$ is defined in Equation 6. A star next to OOS-$\overline{R}^2$ is based on significance of MSE- (2004), which tests for equal MSE of the unconditional forecast and the conditional forecast. One-sided critical values of MSE statistics are bootstrapped distributions, except for caya and all models where they are obtained from McCracken (2004). Critical values for the ms model is calculated as the fraction of draws where the simulated ΔRMSE is greater than the empirically calculated 95% critical value. The two column are for all simulations and for those simulations in which the in-sample estimate was significant at the 95% level. Significance levels a denoted by one, two, and three stars, respectively.**

| | | | Full Sample | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Forecasts begin 20 years after sample | | | | | Forecasts begin 1965 | |
| | | | IS | IS for | OOS | | | IS for | OOS | |
| | Variable | Data | $\overline{R}^2$ | OOS $\overline{R}^2$ | $\overline{R}^2$ | ΔRMSE | Power | OOS $\overline{R}^2$ | $\overline{R}^2$ | ΔRMSE |
| **Full Sample, Not Significant IS** | | | | | | | | | | |
| **dfy** | Default yield spread | 1919–2005 | −1.18 | | −3.29 | −0.14 | | | −4.15 | −0.12 |
| **infl** | Inflation | 1919–2005 | −1.00 | | −4.07 | −0.20 | | | −3.56 | −0.08 |
| **svar** | Stock variance | 1885–2005 | −0.76 | | −27.14 | −2.33 | | | −2.44 | +0.01 |
| **d/e** | Dividend payout ratio | 1872–2005 | −0.75 | | −4.33 | −0.31 | | | −4.99 | −0.18 |
| **lty** | Long term yield | 1919–2005 | −0.63 | | −7.72 | −0.47 | | | −12.57 | −0.76 |
| **tms** | Term spread | 1920–2005 | 0.16 | | −2.42 | −0.07 | | | −2.96 | −0.03 |
| **tbl** | Treasury-bill rate | 1920–2005 | 0.34 | | −3.37 | −0.14 | | | −4.90 | −0.18 |
| **dfr** | Default return spread | 1926–2005 | 0.40 | | −2.16 | −0.03 | | | −2.82 | −0.02 |
| **d/p** | Dividend price ratio | 1872–2005 | 0.49 | | −2.06 | −0.11 | | | −3.69 | −0.09 |
| **d/y** | Dividend yield | 1872–2005 | 0.91 | | −1.93 | −0.10 | | | −6.68 | −0.31 |
| **ltr** | Long term return | 1926–2005 | 0.99 | | −11.79 | −0.76 | | | −18.38 | −1.18 |
| **e/p** | Earning price ratio | 1872–2005 | 1.08 | | −1.78 | −0.08 | | | −1.10 | 0.11 |

**Table 1**
**Continued**

| | | | Full Sample | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Forecasts begin 20 years after sample | | | | | Forecasts begin 1965 | |
| | | | IS | IS for | OOS | | | IS for | OOS | |
| | Variable | Data | $\overline{R}^2$ | OOS $\overline{R}^2$ | $\overline{R}^2$ | ΔRMSE | Power | OOS $\overline{R}^2$ | $\overline{R}^2$ | ΔRMSE |
| **Full Sample, Significant IS** | | | | | | | | | | |
| **b/m** | Book to market | 1921–2005 | **3.20*** | 1.13 | −1.72 | −0.01 | 42 (67) | −7.29 | −12.71 | −0.77 |
| **i/k** | Invstmnt capital ratio | 1947–2005 | **6.63**** | −0.25 | −1.77 | 0.07 | 47 (77) | | Same | |
| **ntis** | Net equity expansion | 1927–2005 | **8.15***** | −4.21 | −5.07 | −0.26 | 57 (78) | 0.96 | −6.79 | −0.32 |
| **eqis** | Pct equity issuing | 1927–2005 | **9.15***** | 2.81 | **2.04**** | 0.30 | 72 (85) | 3.64 | −1.00 | 0.12 |
| **all** | Kitchen sink | 1927–2005 | **13.81**** | 2.62 | −139.03 | −5.97 | - (-) | −20.91 | −176.18 | −6.19 |
| **Full sample, no IS equivalent (caya, ms) or Ex-Post Information (cayp)** | | | | | | | | | | |
| **cayp** | Cnsmptn, wlth, incme | 1945–2005 | **15.72***** | 20.70 | **16.78***** | 1.61 | - (-) | | Same | |
| **caya** | Cnsmptn, wlth, incme | 1945–2005 | - | - | −4.33 | −0.14 | - (-) | | Same | |
| **ms** | Model selection | 1927–2005 | - | - | −22.50 | −1.69 | - (-) | - | −23.71 | −1.79 |
| **1927-2005 Sample, Significant IS** | | | | | | | | | | |
| **d/y** | Dividend yield | 1927–2005 | **2.71*** | | | | | −0.35 | −6.44 | −0.30 |
| **e/p** | Earning price ratio | 1927–2005 | **3.20*** | | | | | −0.94 | −3.15 | −0.05 |
| **b/m** | Book to market | 1927–2005 | **4.14*** | | | | | −8.65 | −19.46 | −1.26 |

## 4.2 Table 2: five-year forecasts

**Read:**

- Five-year overlapping forecasts give some variables larger in-sample $R^2$ values because long-horizon returns are smoother and more persistent.
- Several valuation ratios and spreads appear stronger at long horizons than at annual horizons.
- OOS performance still remains unstable. Many OOS $R^2$ and $\Delta RMSE$ entries are negative, especially in the post-1965 forecast windows.
- The consumption-wealth-income variable with ex-post construction can look strong, but this is not a usable real-time forecast.
- Model selection performs disastrously in the five-year experiment, again showing the cost of searching across many models.

**Economic message:** Long-horizon predictability is not robust enough to rescue the case for aggregate return predictability. Smoother horizons can make in-sample fit look better without making real-time forecasts reliable.

**Table 2**
**Forecasts at 5-year frequency This table is identical to Table 1, except that we predict overlapping 5-yearly equity premia, rather than annual equity premi**

| | | | Full sample | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Forecasts begin 20 years after sample | | | | | | Forecasts begin 1965 | | |
| | | | IS | IS for | OOS | | | IS for | OOS | | |
| | Variable | Data | $\overline{R}^2$ | OOS $\overline{R}^2$ | $\overline{R}^2$ | $\Delta$RMSE | Power | OOS $\overline{R}^2$ | $\overline{R}^2$ | $\Delta$RMSE | |
| **Full Sample, Not Significant IS** | | | | | | | | | | | |
| **ltr** | Long term return | 1926–2005 | −1.36 | | −7.40 | −1.10 | | | −18.92 | −2.72 | |
| **dfr** | Default return spread | 1926–2005 | −1.36 | | −5.71 | −0.77 | | | −4.01 | −0.25 | |
| **infl** | Inflation | 1919–2005 | −1.21 | | −11.25 | −1.70 | | | −7.34 | −0.85 | |
| **lty** | Long term yield | 1919–2005 | −0.15 | | −122.13 | −17.41 | | | −72.47 | −10.96 | |
| **svar** | Stock variance | 1885–2005 | 0.33 | | −79.33 | −13.31 | | | −2.37 | 0.03 | |
| **d/e** | Dividend payout ratio | 1872–2005 | 0.66 | | −4.87 | −0.76 | | | −0.64 | 0.31 | |
| **dfy** | Default yield spread | 1919–2005 | 3.54 | | −59.33 | −9.18 | | | 4.97* | +1.38 | |
| **tbl** | Treasury-bill rate | 1920–2005 | 3.83 | | −17.66 | −2.78 | | | −30.19 | −4.70 | |
| **d/y** | Dividend yield | 1872–2005 | 6.04 | | −4.45 | −0.68 | | | −16.84 | −2.19 | |
| **e/p** | Earning price ratio | 1872–2005 | 6.24 | | −1.04 | −0.03 | | | −3.03 | −0.07 | |
| **tms** | Term spread | 1920–2005 | 7.84 | | −26.52 | −4.24 | | | **10.46**** | +2.44 | |
| **eqis** | Pct equity issuing | 1927–2005 | 9.50 | | −2.35 | −0.11 | | | −5.75 | −0.56 | |
| **b/m** | Book to market | 1921–2005 | 10.78 | | −13.06 | −2.03 | | | −46.34 | −7.17 | |
| **Full Sample, Significant IS** | | | | | | | | | | | |
| **ntis** | Net equity expansion | 1927–2005 | **6.59*** | −8.28 | −3.46 | −0.32 | 21(70) | 1.49 | −13.77 | −1.92 | 2 |
| **d/p** | Dividend price ratio | 1872–2005 | **10.24*** | 14.35 | −1.19* | −0.06 | 21(69) | 8.30 | −26.09 | −3.54 | 2 |
| **i/k** | Invstmnt capital ratio | 1947–2005 | **33.99***** | 27.42 | **12.99**** | +3.39 | 22(78) | | Same | | |
| **all** | Kitchen sink | 1927–2005 | **41.48***** | 43.29 | −499.83 | −45.47 | −(−) | 19.75 | −442.08 | −34.19 | |

**Table 2**
**(Continued)**

| | | | Full sample | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Forecasts begin 20 years after sample | | | | | | Forecasts begin 1965 | | |
| | | | IS | IS for | OOS | | | IS for | OOS | | |
| | Variable | Data | $\overline{R}^2$ | OOS $\overline{R}^2$ | $\overline{R}^2$ | $\Delta$RMSE | Power | OOS $\overline{R}^2$ | $\overline{R}^2$ | $\Delta$RMSE | |
| **Full Sample, No IS Equivalent (caya, ms) or Ex-Post Information (cayp)** | | | | | | | | | | | |
| **cayp** | Cnsmptn, wlth, incme | 1945–2005 | **36.05***** | 63.11 | **30.35***** | +7.50 | - (-) | | | Same | |
| **caya** | Cnsmptn, Wlth, Incme | 1945–2005 | – | – | **9.10***** | +2.50 | - (-) | | | Same | |
| **ms** | Model Selection | 1927–2005 | – | – | −14465.67 | −408.06 | - (-) | – | −122.89 | −18.03 | - |
| **1927-2005 Sample, Significant IS** | | | | | | | | | | | |
| **tms** | Term Spread | 1927–2005 | **12.47*** | | | | | 23.24 | **12.59**** | +2.77 | 11 |
| **e/p** | Earning Price Ratio | 1927–2005 | **14.96*** | | | | | −4.04 | −15.33 | −2.18 | 28 |
| **d/y** | Dividend Yield | 1927–2005 | **14.99*** | | | | | 6.16 | −9.47 | −1.19 | 22 |
| **d/p** | Dividend Price Ratio | 1927–2005 | **21.24**** | | | | | 4.28 | −12.69 | −1.74 | 25 |

## 4.3 Table 3 and Figure 3: monthly forecasts with restrictions

**Read: Table 3:**

- The table evaluates monthly forecasts using the Campbell–Thompson procedure.
- Some variables obtain positive monthly OOS $R^2$ values after restrictions, including interest-rate variables, cross-sectional premium, percent equity issuing, and some valuation ratios.
- Certainty-equivalent utility gains are often small and sensitive to position restrictions.
- Several variables that are significant in sample still have negative OOS performance.

**Read: Figure 3:**

- The monthly plots show three lines: the raw OOS prediction, the Campbell–Thompson truncated prediction, and the plain prediction.
- Restrictions can stabilize forecasts, but the cumulative performance remains highly time varying.
- For some predictors, performance improves only in specific periods and deteriorates later.

**Economic message:** Monthly restrictions help some predictors, but they do not provide a broad, stable, economically large solution to the equity premium prediction problem.

1484

**Table 3**
**Forecasts at monthly frequency using Campbell and Thompson (2005) procedure**
**Refer to Table 1 for basic explanations. This table presents statistics on forecast errors in-sample (IS) and out-of-sample (OOS) for equity premium forecas (both in the forecasting equation and forecast). Variables are explained in Section 2. The data period is December 1927 to December 2004, except for csp (N and cay3 (December 1951 to December 2004). Critical values of all statistics are obtained empirically from bootstrapped distributions, except for cay3 model McCracken (2004). The resulting significance levels at 90%, 95%, and 99% are denoted by one, two, and three stars, respectively. They are two-sided for IS mod for OOS superior model performance. The first data column is the IS $\overline{R}^2$ when returns are logged, as they are in our other tables. The remaining columns a returns for correspondence with Campbell and Thompson (2005). Certainty Equivalence (CEV) gains are based on the utility of an optimizer with a risk-aver trades based on unconditional forecast and conditional forecast. Equity positions are winsorized at 150% ($w = w_{max}$). At this risk-aversion, the base CEV are 8 on the unconditional forecast, 79bp for the market, and 40bp for the risk-free rate. "T" means "truncated" to avoid a negative equity premium prediction. "U" to avoid a forecast that is based on a coefficient that is inverse to what the theory predicts. A superscript *h* denotes high trading turnover of about 10%/month n based on unconditional forecasts.**

| | | Log returns | Simple returns | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | IS | | OOS | Campbell and Thompson (2005) OOS | | | | |
| | Variable | IS $\overline{R}^2$ | $\overline{R}^2$ | $\overline{R}^2$ T | $\overline{R}^2$ | Frcst= T | Frcst= U | $\overline{R}^2$ TU | ΔRMSE TU | $w = w_{max}$ |
| **d/e** | Dividend payout ratio | 0.02 | −0.10 | −0.10 | −0.70 | 0.0 | 7.9 | −0.69 | −0.0114 | 57.7 |
| **svar** | Stock variance | −0.09 | −0.07 | −0.07 | −0.79 | 0.0 | 0.0 | −0.79 | −0.0134 | 35.4 |
| **dfr** | Default return spread | −0.02 | −0.07 | −0.08 | −0.37 | 0.0 | 20.9 | −0.29 | −0.0030 | 44.5 |
| **lty** | Long term yield | −0.03 | 0.02 | 0.02 | −0.80 | 34.1 | 0.0 | **0.26**** | +0.0085 | 19.5 |
| **ltr** | Long term return | 0.04 | 0.07 | 0.08 | −0.63 | 3.0 | 38.2 | **0.11**** | +0.0053 | 51.2 |
| **infl** | Inflation | −0.01 | 0.14 | −0.05 | **0.01*** | 1.3 | 0.0 | **0.07**** | +0.0045 | 43.5 |
| **tms** | Term spread | 0.12 | 0.18 | 0.20 | **0.22**** | 3.7 | 0.0 | **0.21**** | +0.0073 | 59.2 |
| **tbl** | Treasury-bill rate | 0.10 | **0.20*** | 0.15 | −0.08* | 23.1 | 0.0 | **0.25**** | +0.0081 | 16.4 |
| **dfy** | Default yield spread | −0.06 | **0.28*** | 0.28 | −0.56 | 4.0 | 0.0 | −0.49 | −0.0071 | 27.3 |
| **d/p** | Dividend price ratio | 0.12 | **0.33*** | 0.29 | −0.30 | 32.3 | 0.0 | **0.17*** | +0.0066 | 16.1 |
| **d/y** | Dividend yield | **0.22*** | **0.47***** | 0.45 | −1.12 | 54.2 | 0.0 | −0.04* | +0.0023 | 16.4 |
| **e/p** | Earning price ratio | **0.51**** | **0.54***** | 0.45 | −1.04 | 18.1 | 0.0 | −1.03 | −0.0183 | 34.4 |
| **eqis** | Pct equity issuing | **0.82***** | **0.80***** | 0.59 | **0.14**** | 6.7 | 0.0 | **0.30***** | +0.0093 | 55.8 |
| **b/m** | Book to market | **0.45**** | **0.81***** | 0.88 | −3.28 | 44.3 | 0.0 | −2.23 | −0.0432 | 31.1 |
| **$e^{10}/p$** | Earning(10Y) price ratio | **0.46***** | **0.86***** | 0.96 | −2.21 | 52.4 | 0.0 | −0.48** | −0.0071 | 15.4 |

Table 3, monthly forecasts using Campbell–Thompson procedures

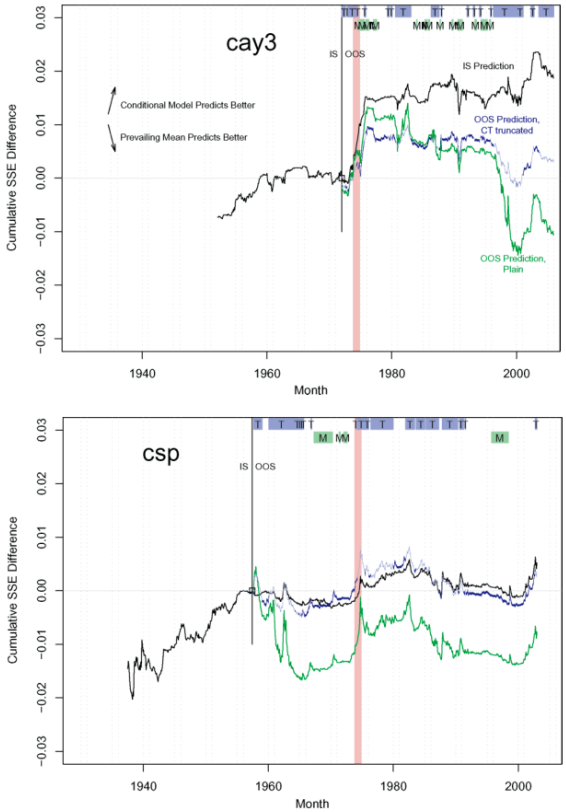



**Figure 3**
**Monthly performance of in-sample significant predictors**
Explanation: These figures are the analogs of Figures 1 and 2, plotting the IS and OOS performance of t named model. However, they use monthly data. The IS performance is in black. The Campbell-Thomps (2005) (CT) OOS model performance is plotted in blue, the plain OOS model performance is plotted green. The top bars ("T") indicate truncation of the equity prediction at 0, inducing the CT investor

Figure 3, panels *cay3* and *csp*

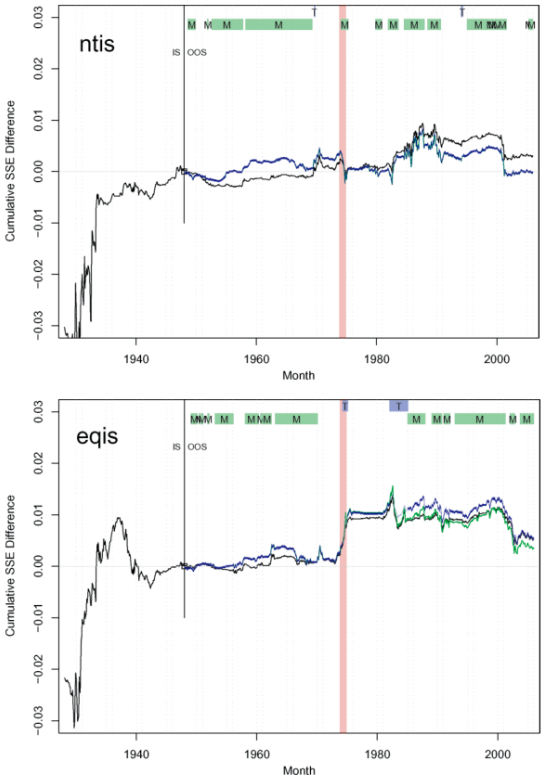



**Figure 3 Continued**

predictions. For example, **d/y**'s equity premium predictions are truncated to zero in 54.2% of all months; **csp**'s predictions are truncated in 44.7% of all months. Truncation is a very effective constraint.

Figure 3, panels *ntis* and *eqis*

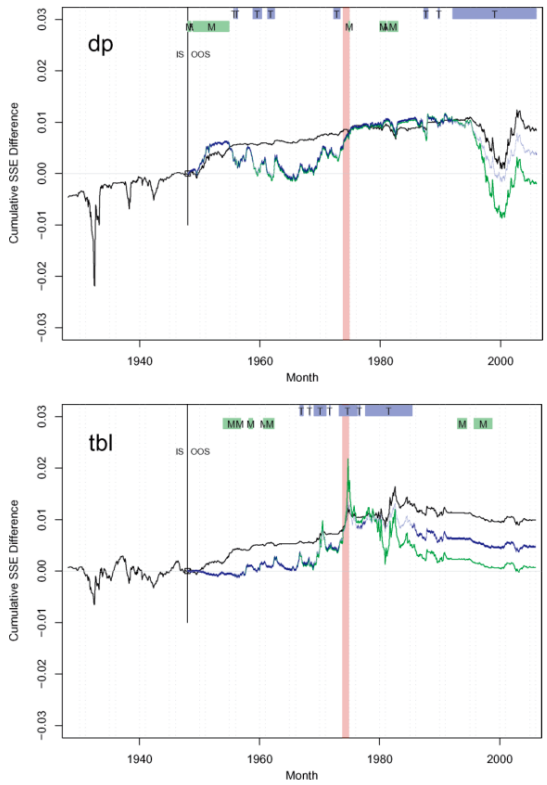



**Figure 3 Continued**

coefficient sign constraint matters only for **dfr** and **ltr** (and mildly for d/
None of these three models has IS performance high enough to make tl
worthwhile to explore further

Figure 3, panels $d/p$ and $tbl$

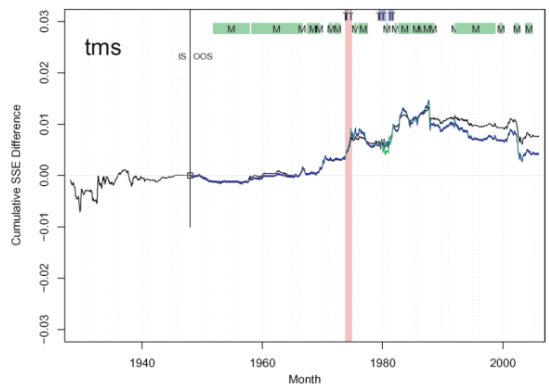



**Figure 3 Continued**

OOS $\overline{R}^2$. This reflects the number of degrees of freedom: even though we have between 400 and 800 data months, the plain $\Delta$RMSE and $R^2$ are

Figure 3, panel $tms$

## 4.4 Tables 4–5: smoothed ratios and persistence corrections

**Read: Table 4:**

- The paper tries smoothed dividend and earnings ratios, including multi-year averages.
- Smoothing can create significant in-sample evidence, especially at annual and five-year horizons.
- Out-of-sample results remain mixed and often negative, especially for forecasts beginning in 1965.

**Read: Table 5:**

- The paper compares ordinary OLS with Stambaugh and Lewellen corrections for persistent regressors.
- These corrections change coefficients and some in-sample statistics.
- They do not generally turn poor OOS forecasts into reliable successful forecasts.

**Economic message:** Robust inference and alternative ratio definitions do not overturn the main result. The issue is not merely a particular estimator or a particular valuation-ratio construction.

**Significant forecasts using various d/p, e/p, and d/e Ratios**

| | Variable | Data | Freq | IS $\overline{R}^2$ | OOS $\overline{R}^2$ | OOS ΔRMSE |
|---|---|---|---|---|---|---|
| **e/p** | Earning(1Y) price ratio | 1927–2005 | M 1965– | **0.54**$^{**}$ | −1.20 | −0.02 |
| **$e^5$/p** | Earning(5Y) price ratio | 1927–2005 | M 1965– | **0.32**$^{*}$ | −0.60 | −0.01 |
| **$e^{10}$/p** | Earning(10Y) price ratio | 1927–2005 | M 1965– | **0.49**$^{**}$ | −0.83 | −0.01 |
| **$e^3$/p** | Earning(3Y) price ratio | 1882–2005 | A 1902– | **2.53**$^{**}$ | −1.05$^{*}$ | −0.01 |
| **$e^5$/p** | Earning(5Y) price ratio | 1882–2005 | A 1902– | **2.88**$^{**}$ | −0.52$^{*}$ | +0.04 |
| **$e^{10}$/p** | Earning(10Y) price ratio | 1882–2005 | A 1902– | **4.89**$^{**}$ | **2.12**$^{**}$ | +0.30 |
| **$d^{10}$/p** | Dividend(3Y) price ratio | 1882–2005 | A 1902– | **1.85**$^{*}$ | −1.53 | −0.05 |
| **$d^5$/p** | Dividend(5Y) price ratio | 1882–2005 | A 1902– | **2.48**$^{*}$ | −0.54$^{*}$ | +0.04 |
| **$d^{10}$/p** | Dividend(10Y) price ratio | 1882–2005 | A 1902– | **2.11**$^{*}$ | −1.07$^{*}$ | −0.01 |
| **$e^3$/p** | Earning(3Y) price ratio | 1882–2005 | A 1965– | **2.53**$^{**}$ | −3.41 | −0.06 |
| **$e^5$/p** | Earning(5Y) price ratio | 1882–2005 | A 1965– | **2.88**$^{**}$ | −5.01 | −0.19 |
| **$e^{10}$/p** | Earning(10Y) price ratio | 1882–2005 | A 1965– | **4.89**$^{**}$ | −11.45 | −0.66 |
| **$d^3$/p** | Dividend(3Y) price ratio | 1882–2005 | A 1965– | **1.85**$^{*}$ | −6.55 | −0.30 |
| **$d^5$/p** | Dividend(5Y) price ratio | 1882–2005 | A 1965– | **2.48**$^{*}$ | −8.79 | −0.47 |
| **$d^{10}$/p** | Dividend(10Y) price ratio | 1882–2005 | A 1965– | **2.11**$^{*}$ | −8.32 | −0.43 |
| **$e^3$/p** | Earning(3Y) price ratio | 1882–2005 | 5Y 1902– | **11.35**$^{*}$ | **3.46**$^{**}$ | +0.89 |
| **$e^5$/p** | Earning(5Y) price ratio | 1882–2005 | 5Y 1902– | **16.16**$^{**}$ | **4.76**$^{**}$ | +1.16 |
| **$e^{10}$/p** | Earning(10Y) price ratio | 1882–2005 | 5Y 1902– | **16.47**$^{**}$ | −2.85$^{*}$ | −0.37 |
| **d/p** | Dividend(1Y) price ratio | 1882–2005 | 5Y 1902– | **12.30**$^{*}$ | −0.66$^{*}$ | +0.06 |
| **$d^3$/p** | Dividend(3Y) price ratio | 1882–2005 | 5Y 1902– | **13.11**$^{*}$ | −2.02$^{*}$ | −0.21 |
| **$d^5$/p** | Dividend(5Y) price ratio | 1882–2005 | 5Y 1902– | **13.75**$^{*}$ | −3.85$^{*}$ | −0.57 |
| **$e^3$/p** | Earning(3Y) price ratio | 1882–2005 | 5Y 1965– | **11.35**$^{*}$ | −12.55 | −1.56 |
| **$e^5$/p** | Earning(5Y) price ratio | 1882–2005 | 5Y 1965– | **16.16**$^{**}$ | −21.16 | −2.85 |
| **$e^{10}$/p** | Earning(10Y) price ratio | 1882–2005 | 5Y 1965– | **16.47**$^{**}$ | −25.65 | −3.51 |
| **d/p** | Dividend(1Y) price ratio | 1882–2005 | 5Y 1965– | **12.30**$^{*}$ | −29.33 | −4.03 |
| **$d^3$/p** | Dividend(3Y) price ratio | 1882–2005 | 5Y 1965– | **13.11**$^{*}$ | −28.11 | −3.86 |
| **$d^5$/p** | Dividend(5Y) price ratio | 1882–2005 | 5Y 1965– | **13.75**$^{*}$ | −30.71 | −4.23 |

Refer to Table 1 for basic explanations. The table reports only those combinations of **d/p**, **e/p**, and **d/e** that were found to predict equity premia significantly in-sample. This table presents statistics on forecast errors in-sample (IS) and out-of-sample (OOS) for excess stock return forecasts at various frequencies. Variables are explained in Section 2. All ΔRMSE numbers are in percent per frequency corresponding to the column entitled 'Freq'. The 'Freq' column also gives the first year of forecast. A star next to OOS-$\overline{R}^2$ is based on the MSE-$F$-statistic by McCracken (2004), which tests for equal MSE of the unconditional forecast and the conditional forecast. One-sided critical values of MSE statistics are obtained empirically from bootstrapped distributions. Significance levels at 90%, 95%, and 99% are denoted by one, two, and three stars, respectively.

Table 4: smoothed dividend and earnings ratios

**Table 5**
**Forecasts at monthly frequency with alternative procedures and total returns**
**Refer to Table 1 for basic explanations.Columns under the heading "OLS" are unadjusted betas, columns under the heading "Stambaugh" correct for betas foll (1999), and columns under the heading 'Lewellen' correct for betas following Lewellen (2004). $\rho$ under the column OLS gives the autoregressive coefficient of th entire sample period (the variables are sorted in descending order of $\rho$).**

| | | | OLS | | | | Stambaugh | | | | Lewe | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | IS | OOS | | IS | OOS | | | IS | |
| | Variable | Data | $\rho$ | $\overline{R}^2$ | $\overline{R}^2$ | Power | $\overline{R}^2$ | $\overline{R}^2$ | Power | | $\overline{R}^2$ | $\overline{R}^2$ |
| **d/e** | Dividend payout ratio | 192701–200512 | 0.9989 | 0.01 | −2.02 | 15 (70) | 0.01 | −2.11 | 15 (69) | | 0.01 | −2. |
| **lty** | Long term yield | 192701–200512 | 0.9963 | −0.01 | −1.15 | 9 (68) | −0.01 | −1.71 | 9 (68) | | −0.01 | −1. |
| **d/y** | Dividend yield | 192701–200512 | 0.9929 | **0.25**$^{*}$ | −0.40 | 33 (71) | **0.25**$^{*}$ | −0.36 | 33 (71) | | **0.25**$^{*}$ | −0. |
| **d/p** | Dividend price ratio | 192701–200512 | 0.9927 | 0.15 | −0.15 | 29 (56) | 0.05 | −0.31 | 26 (69) | | −0.15$^{**}$ | −1. |
| **tbl** | Treasury-bill rate | 192701–200512 | 0.9922 | 0.11 | −0.18 | 19 (69) | 0.11 | −0.33 | 20 (69) | | 0.11 | −0. |
| **e/p** | Earning price ratio | 192701–200512 | 0.9879 | **0.54**$^{**}$ | −1.21 | 56 (64) | **0.48**$^{**}$ | −0.54 | 59 (73) | | **0.02**$^{***}$ | −0. |
| **b/m** | Book to market | 192701–200512 | 0.9843 | **0.40**$^{**}$ | −2.45 | 48 (65) | **0.36**$^{**}$ | −1.61 | 48 (71) | | −0.14$^{**}$ | −0. |
| **csp** | Cross-Sectional prem | 193705–200212 | 0.9788 | **0.92**$^{***}$ | **0.70**$^{***}$ | 65 (80) | **0.92**$^{***}$ | **0.70**$^{***}$ | 65 (80) | | **0.91**$^{***}$ | **0.** |
| **dfy** | Default yield spread | 192701–200512 | 0.9763 | −0.07 | −0.14 | 9 (59) | −0.07 | −0.33 | 8 (59) | | −0.15 | −0. |
| **ntis** | Net equity expansion | 192701–200512 | 0.9680 | **0.75**$^{***}$ | −0.28 | 59 (76) | **0.75**$^{***}$ | −0.29 | 59 (76) | | **0.74**$^{***}$ | −0. |
| **tms** | Term spread | 192701–200512 | 0.9566 | 0.07 | **0.09**$^{*}$ | 21 (66) | 0.07 | **0.07**$^{*}$ | 20 (65) | | 0.07 | **0.** |
| **svar** | Stock variance | 192701–200512 | 0.6008 | −0.08 | −0.34 | 7 (53) | −0.08 | −0.34 | 7 (53) | | −1.66$^{**}$ | −0. |
| **infl** | Inflation | 192701–200512 | 0.5513 | −0.00 | −0.07 | 14 (62) | −0.00 | −0.07 | 14 (62) | | −0.03 | −0. |
| **ltr** | Long term return | 192701–200512 | 0.0532 | 0.04 | −0.49 | 18 (62) | 0.04 | −0.48 | 18 (62) | | −1.55$^{***}$ | −6. |
| **dfr** | Default return spread | 192701–200512 | −0.1996 | −0.02 | −0.30 | 12 (61) | −0.02 | −0.30 | 12 (61) | | −1.32$^{*}$ | −2. |

Table 5: persistence corrections and total returns

## 4.5 Table 6: encompassing tests

**Read:**

- Encompassing tests ask whether a forecast combination of the historical mean and a conditional model improves MSE.
- The ex-post optimal weight $\lambda$ can sometimes be positive and significant.
- The rolling version of the combination is much less successful, because the optimal weight itself must be learned in real time.
- Monthly data show many significant encompassing statistics, but rolling $\Delta RMSE$ values are often negative or small.

**Economic message:** Ex-post combinations can make models look useful, but real-time combinations are difficult. Learning which predictor to trust is itself a hard forecasting problem.

# 5 Short summary

- Welch and Goyal reexamine the empirical performance of equity premium predictors proposed in the literature.
- They evaluate predictors under common data definitions, common forecasting windows, and common inference procedures.
- The benchmark is the historical average equity premium.
- The main predictive regression is a linear regression of the equity premium on a lagged predictor.

**Table 6**
**Encompassing tests**
**This table presents statistics on encompassing tests for excess stock return forecasts at various frequencies. Variables are explained in Section 1. All numbers are in**
**corresponding to the panel. λ gives the ex-post weight on the conditional forecast for the optimal forecast that minimizes the MSE. ENC is the test statistic pr**
**McCracken (2001) for a test of forecast encompassing. One-sided critical values of ENC statistic are obtained empirically from bootstrapped distributions, except f**
**models where they are obtained from Clark and McCracken (2001). Critical values for ms model are not calculated. cayp uses ex-post information. ΔRMSE* is**
**between the unconditional forecast and the optimal forecast for the same sample/forecast period. ΔRMSE$^{ef}$ is the RMSE difference between the unconditional for**
**forecast for the same sample/forecast period using rolling estimates of λ. Significance levels at 90%, 95%, and 99% are denoted by one, two, and three stars, respectiv**

| | | Estimation: | All data | | | | | All Data | | | | | | After 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | OOS Forecast: | After 20 years | | | | | After 1965 | | | | | | After 1 |
| | | Data | $\overline{R}^2$ | λ | ENC | ΔRMSE* | ΔRMSE$^{ef}$ | λ | ENC | ΔRMSE* | ΔRMSE$^{ef}$ | $\overline{R}^2$ | λ | ENC |
| **d/p** | Dividend Price Ratio | 1872–2005 | 0.49 | 0.21 | 0.48 | +0.0084 | −0.2583 | 0.40 | **0.87**$^{*}$ | +0.0664 | −0.4989 | 1.67 | 0.54 | **2.19**$^{**}$ |
| **d/y** | Dividend Yield | 1872–2005 | 0.91 | 0.38 | 1.94 | +0.0614 | −0.5713 | 0.30 | **1.24**$^{*}$ | +0.0749 | −0.5389 | **2.71**$^{*}$ | 0.41 | **3.24**$^{**}$ |
| **e/p** | Earning price ratio | 1872–2005 | 1.08 | 0.22 | 0.40 | +0.0074 | −0.2266 | 0.66 | **1.21**$^{**}$ | +0.1508 | −0.4845 | **3.20**$^{*}$ | 0.48 | **2.51**$^{**}$ |
| **d/e** | Dividend payout ratio | 1872–2005 | −0.75 | −1.73 | −1.46 | +0.2135 | +0.0960 | −8.46 | −0.45 | +0.7545 | +0.2858 | −1.24 | −4.57 | −1.25 |
| **svar** | Stock variance | 1885–2005 | −0.76 | −0.42 | −4.74 | +0.2387 | −0.6475 | 2.07 | 0.03 | +0.0134 | −0.5937 | −1.32 | −16.73 | −0.18 |
| **b/m** | Book to market | 1921–2005 | **3.20**$^{*}$ | 0.49 | **4.16**$^{**}$ | +0.2532 | −0.0575 | 0.20 | **1.27**$^{*}$ | +0.0559 | −0.7885 | **4.14**$^{*}$ | 0.18 | **1.67**$^{*}$ |
| **ntis** | Net equity expansion | 1927–2005 | **8.15**$^{***}$ | 0.31 | 1.46 | +0.0619 | −0.2708 | 0.31 | **1.30**$^{*}$ | +0.0805 | −0.9310 | **8.15**$^{***}$ | 0.31 | **1.30**$^{*}$ |
| **eqis** | Pet equity issuing | 1927–2005 | **9.15**$^{***}$ | 0.67 | **4.45**$^{**}$ | +0.3917 | −0.0564 | 0.56 | **3.12**$^{**}$ | +0.3342 | −0.7106 | **9.15**$^{***}$ | 0.56 | **3.12**$^{**}$ |
| **tbl** | Treasury-bill rate | 1920–2005 | 0.34 | 0.39 | **2.14**$^{*}$ | +0.1031 | −1.2425 | 0.41 | **2.16**$^{**}$ | +0.1790 | −1.3058 | 0.15 | 0.33 | **2.72**$^{**}$ |
| **lty** | Long term yield | 1919–2005 | −0.63 | 0.29 | **2.67**$^{*}$ | +0.0971 | −0.7012 | 0.28 | **2.39**$^{**}$ | +0.1447 | −0.9358 | −0.94 | 0.25 | **2.39**$^{**}$ |
| **ltr** | Long term return | 1926–2005 | 0.99 | 0.31 | **4.55**$^{**}$ | +0.2077 | −0.1412 | 0.24 | **2.45**$^{**}$ | +0.1300 | −8.4290 | 0.92 | 0.25 | **2.44**$^{**}$ |
| **tms** | Term Spread | 1920–2005 | 0.16 | 0.38 | 0.93 | +0.0433 | −1.0292 | 0.47 | **1.07**$^{*}$ | +0.0977 | −0.8750 | 0.89 | 0.50 | **1.95**$^{**}$ |
| **dfy** | Default yield spread | 1919–2005 | −4.18 | −2.62 | −0.48 | +0.1503 | −0.9718 | −10.65 | −0.30 | +0.6395 | +0.4999 | −1.31 | −11.91 | −0.24 |
| **dfr** | Default Return Spread | 1926–2005 | 0.40 | 0.44 | 0.87 | +0.0501 | −0.3698 | 0.47 | 0.78 | +0.0710 | −0.3808 | 0.32 | 0.48 | 0.74 |
| **infl** | Inflation | 1919–2005 | −1.00 | −2.46 | −0.68 | +0.2019 | −0.4520 | −1.48 | −0.15 | +0.0429 | −15.1368 | −0.99 | −3.12 | −0.86 |
| **ik** | Invstmnt capital ratio | 1947–2005 | **6.63**$^{**}$ | 0.53 | **3.01**$^{**}$ | +0.3330 | −0.1950 | 0.53 | **3.01**$^{**}$ | +0.3330 | −0.1950 | **6.63**$^{**}$ | 0.53 | **3.01**$^{**}$ |
| **cayp** | Cnsmptn, wlth, incme | 1945–2005 | **15.72**$^{***}$ | 1.34 | **7.62**$^{***}$ | +1.7225 | +0.3315 | 1.34 | **7.62**$^{***}$ | +1.7225 | +0.3315 | **15.72**$^{***}$ | 1.34 | **7.62**$^{***}$ |
| **all** | Kitchen sink | 1927–2005 | **13.81**$^{**}$ | 0.13 | 4.86 | +0.1607 | +0.0160 | −0.07 | −1.26 | +0.0342 | −0.4666 | **13.81**$^{**}$ | −0.07 | −1.26 |
| **caya** | Cnsmptn, wlth, incme | 1945–2005 | - | 0.45 | **3.39**$^{**}$ | +0.3117 | −0.3185 | 0.45 | **3.39**$^{**}$ | +0.3117 | −0.3185 | - | 0.45 | **3.39**$^{**}$ |
| **ms** | Model Selection | 1927–2005 | - | 0.24 | 4.82 | +0.1870 | +0.0739 | 0.07 | 0.59 | +0.0094 | −1.1268 | - | 0.07 | 0.59 |

**Table 6**
**panel B: Monthly Data**

| | | OOS Forecast: | After 194701 | | | | | | | After 19 |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Data | $\overline{R}^2$ | λ | ENC | ΔRMSE* | ΔRMSE$^{ef}$ | λ | ENC | ΔR |
| **d/p** | Dividend price ratio | 192701–200512 | 0.15 | 0.53 | **4.14**$^{**}$ | +0.0065 | −0.0134 | 0.53 | **2.67**$^{**}$ | +( |
| **d/y** | Dividend yield | 192701–200512 | **0.25**$^{*}$ | 0.43 | **6.53**$^{***}$ | +0.0083 | −0.0115 | 0.45 | **3.90**$^{**}$ | +( |
| **e/p** | Earning price ratio | 192701–200512 | **0.54**$^{**}$ | 0.35 | **9.27**$^{***}$ | +0.0097 | −0.0135 | 0.28 | **3.08**$^{**}$ | +( |
| **d/e** | Dividend payout ratio | 192701–200512 | 0.01 | −0.02 | −0.22 | +0.0000 | −0.0146 | −1.12 | −3.01 | +( |
| **svar** | Stock variance | 192701–200512 | −0.08 | −12.30 | −0.47 | +0.0172 | +0.0046 | −12.93 | −0.32 | +( |
| **csp** | Cross-sectional prem | 193705–200212 | **0.92**$^{***}$ | 0.38 | **6.21**$^{***}$ | +0.0093 | −0.0138 | 0.82 | **5.50**$^{***}$ | +( |
| **b/m** | Book to market | 192701–200512 | **0.40**$^{**}$ | 0.18 | **3.04**$^{**}$ | +0.0016 | −0.0416 | 0.07 | 0.89 | +( |
| **ntis** | Net equity expansion | 192701–200512 | **0.75**$^{***}$ | 0.60 | **4.28**$^{**}$ | +0.0075 | −0.0055 | 0.47 | **2.77**$^{**}$ | +( |
| **tbl** | Treasury-bill rate | 192701–200512 | 0.11 | 0.50 | **5.47**$^{**}$ | +0.0081 | −0.0222 | 0.51 | **4.86**$^{***}$ | +( |
| **lty** | Long term yield | 192701–200512 | −0.01 | 0.35 | **7.57**$^{***}$ | +0.0079 | −0.0084 | 0.35 | **5.47**$^{***}$ | +( |
| **ltr** | Long term return | 192701–200512 | 0.04 | −0.15 | −0.77 | +0.0003 | −4.0129 | 0.30 | **1.02**$^{*}$ | +( |
| **tms** | Term spread | 192701–200512 | 0.07 | 0.68 | **2.51**$^{**}$ | +0.0050 | −0.0311 | 0.73 | **2.37**$^{**}$ | +( |
| **dfy** | Default yield spread | 192701–200512 | −0.07 | −1.04 | −0.27 | +0.0008 | −0.0070 | 2.15 | 0.20 | +( |
| **dfr** | Default return spread | 192701–200512 | −0.02 | −0.85 | −0.72 | +0.0018 | −0.0134 | −0.03 | −0.01 | +( |
| **infl** | Inflation | 192701–200512 | −0.00 | 1.01 | 0.69 | +0.0021 | −0.0114 | 1.19 | 0.58 | +( |
| **all** | Kitchen sink | 192701–200512 | **1.98**$^{***}$ | 0.05 | 4.39 | +0.0008 | −0.0150 | 0.14 | **5.88**$^{**}$ | +( |
| **ms** | Model selection | 192701–200512 | – | 0.09 | 1.51 | +0.0004 | −0.0232 | 0.14 | 1.39 | +( |

- In-sample significance is not enough; the paper emphasizes recursive out-of-sample forecasting.
- Most annual predictors fail out of sample.
- Some predictors are significant only because of early sample periods or the Oil Shock episode.
- Five-year horizons produce some stronger in-sample patterns but still do not deliver robust OOS forecasting success.
- Monthly Campbell–Thompson restrictions improve some forecasts but do not rescue broad predictability.
- Stambaugh and Lewellen corrections do not overturn the central conclusion.
- Kitchen-sink and model-selection approaches often overfit and perform very badly OOS.
- Ex-post forecast combinations can help, but real-time rolling combinations are difficult.
- The overall message is that equity premium prediction is unstable, fragile, and not reliably useful for market timing.

# 6 Conclusion

## 6.1 Core conclusion

Welch and Goyal conclude that the empirical record for aggregate equity premium prediction is weak. A large set of published predictors performs poorly out of sample, and many variables that appear significant in sample are unstable or episode-dependent.

## 6.2 Economic intuition

Expected returns may vary over time, but estimating that variation is difficult because aggregate stock returns are noisy and samples are short. Persistent predictors can look powerful in hindsight while still failing for real-time investors. The historical mean is a simple benchmark, but it is hard to beat robustly.

## 6.3 Takeaway

The paper is a methodological warning for asset pricing. Predictive variables should be judged by real-time OOS performance, not only by in-sample fit. The burden of proof is high: a useful equity premium predictor must be stable, economically meaningful, and better than the historical average when evaluated using only information available at the time.